



University Department of Computer Science

University of Mumbai

A Final Year Internship Report

On

Infrastructure & Cloud Engineer

At

S&L India IT Consulting Services Private Limited

BY

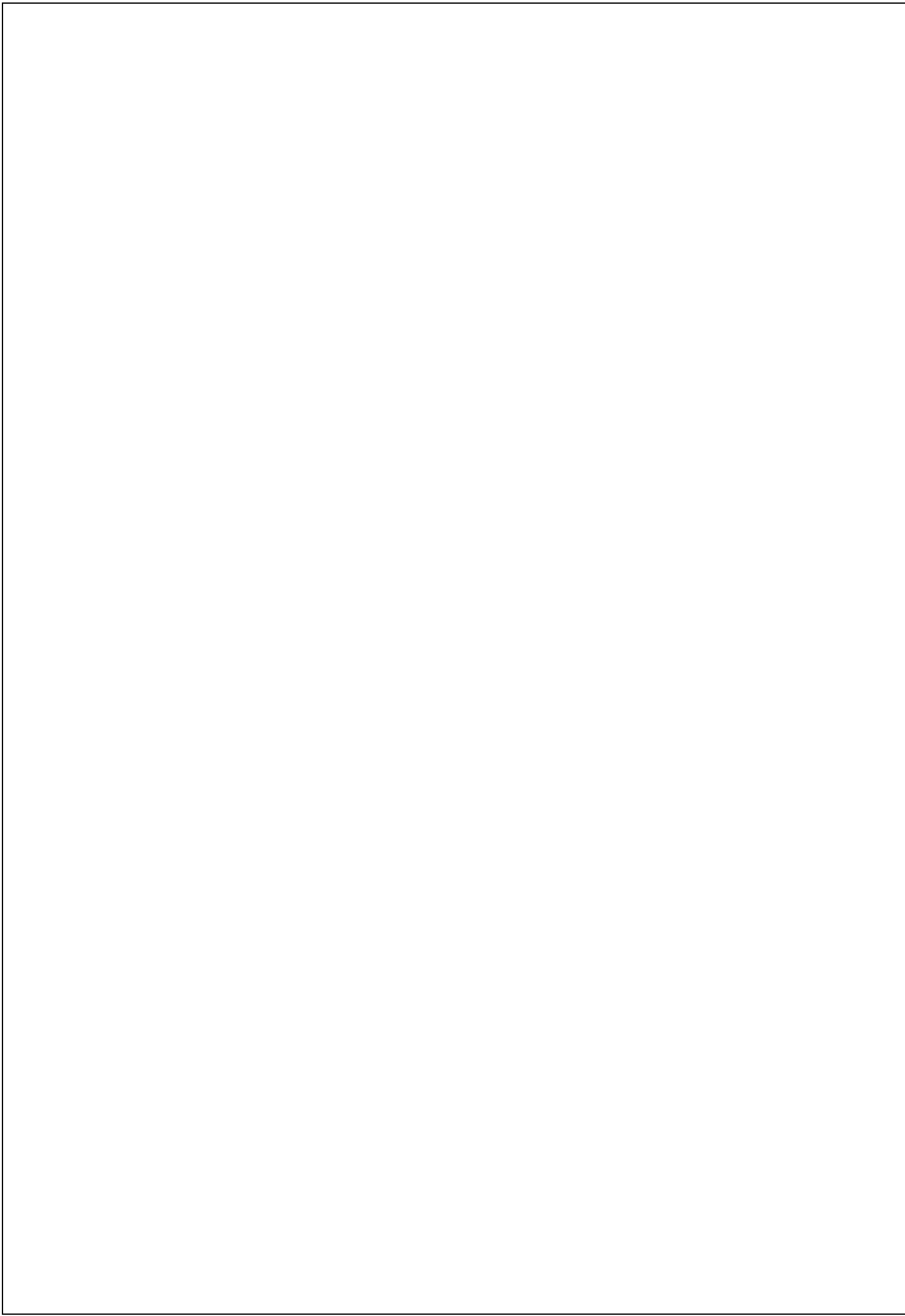
Sharad Dinkar Gore

UDCS406

SUBMITTED TO

UNIVERSITY OF MUMBAI

DEPARTMENT OF COMPUTER SCIENCE (UDCS)



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Mumbai-400098

Appendix 1

This is to certify that Mr. Sharad Dinkar Gore of
University Department of Computer Science (UDCS) College/Institution
worked as an intern as part of her MSc course in Computer Science of
University of Mumbai. The particulars of internship are given below:

Internship starting date: 20/01/2020

Internship ending date: 21/03/2020

Actual number of days worked: 61 Days

Tentative number of hours worked: 1464 Hours

Broad area of work: Managing Infrastructure

A small description of work done by the intern during the period:

- I was responsible for handling client infrastructure in cloud (In Azure and AWS).
- I was responsible to keep production environment highly available
- Monitoring the client environment using various monitoring tool like azure monitor & Site 24x7



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Services

1) NETWORK INFRASTRUCTURE:

- Systems architecture analysis
- Optimized architecture
- On-Premise, Hybrid & Cloud solutions
- Data center virtualization – VMware & Microsoft solutions
- Virtual desktop infrastructure
- Microsoft server solutions and Azure
- Active Directory, Exchange & SQL, and more
- Microsoft Office 365 & Azure solutions
- Server consolidation.

2) SYSTEM ARCHITECTUR :

- Network design optimization
- Data center architecture
- Remote access & VPN
- Software defined networking
- WAN acceleration
- Power & cooling analysis

3) DATA SECURITY:

- Identity Management
- Security gateway
- Internet & Email content filtering
- Advanced Endpoint malware protection
- Vulnerability management
- Cloud security solutions

4) DATA PROTECTION:

- Business continuity planning
- Back-up, recovery processes & procedures
- Co-Location, Cloud or Hybrid solutions
- Server clustering & replication
- Distributed file systems
- Global server load balancing
- Email archiving & retention

Partners

- Amazon web services
- Microsoft Azure
- Google Cloud Platform
- IBM
- DELL
- VMWARE

INTRODUCTION

Internships are formal programs designed to provide practical experience in real world environment to students who are new to that field. Internships help build competent résumé by giving students visible work experience. Although colleges and universities assist students in finding the right Internship programs, it is the responsibility of the would-be interns to carefully examine internship programs and see if those programs actually offer the training(s) they need.

There is no standardized duration for how long an Internship program lasts but typically an Internship is the period of three to twelve months. During this period, interns have the opportunity to explore their field of interest, find out what future they are diving into and whether or not their skill sets match their path of career

PURPOSE OF INTERNSHIP

Internship program provides a new perspective to the students. It provides learning opportunities outside the boundaries of the course curriculum and classroom activities. These opportunities enhance student's capability to apply theoretical understanding into real-world scenarios, thereby enhancing the students' academic and career goals

From the student perspective, internship assists with career development in different diverse fields such as coding, programming, networking, marketing, etc. by providing vital work experience that allows students to explore their area of interests and develop relevant skills and competencies.

From the organization perspective, internship provides a unique opportunity to train fresh talents, enhance their professional development as well as aid the

professional growth of the intern mentor/supervisor. Organizations can also find potential employees within the interns.

OBJECTIVE OF INTERNSHIP

The purpose of this internship is to expose students to real work of environment experience and at the same time. The report is prepared for the purpose of explaining the achievement of the author during internship period. The objectives of internship program are as follows:

- To build the strength, teamwork spirit and self-confidence in students life.
- Provides the opportunity to learn real-life work skills and etiquette hands-on at a real job.
- To Gain valuable skills, knowledge and experience in a field to allow you to make a career transition.
- Explore a career interest, develop skills, and gain experience.
- Internship focus on the skills you will learn and the experiences you will have, not simply on the company name.
- To help you apply the theory and skills you have learned in the classroom setting, and for your personal development.
- The student will be able instilled with good moral values such as responsibility, commitment and trustworthy during their training
- Internships strengthen resumes by giving students visible work experience and providing older job hunters with needed credentials to change careers.

Description about the activities done:

1) Monitoring the client environment using various monitoring tool like azure monitor & Site 24x7.

2) Send daily backup report to client and details information about the backup like overall status its health and confirm that all backup has been taken and informed the same to the client.

If there is any error or health of backup is failed then we have to inform to the client and take all necessary action to sort of the problem.

This task is very important so we have to solve this as soon as possible.

3) check the mail 24*7 if any request done by the clients then we have to process it and inform to the client and give them status about the request raised by the clients. There are many request cant be raised by the clients like below

1) reset the password of user. In this case we have to reset the password of user by taking the remote access of client for that we use bombgar platform to get their access.

After resetting the password we will share it only with requested user.

2) Migration request : in this case we have to migrate the user from client side to office365 to get him access to this account from any device he wants.

There are many steps include in this migration first we have to ask them to give access to the account then we start the processes with sync to that account .we have to update them so after completing the synced we move to next step that is run the command in PowerShell to enable their account.

This process will take 1 hour to get done but some time it also depend on the server and number of request at point.

4) monitoring the Azure 24*7 and its all working and check whether all Vm are running or not. we have to check all VM and confirm its working fine and taking backup .if any vm stop working then we have to take action and try to solve as soon as possible.

Below software we use for monitoring purpose:

- 1) Cloudally
- 2) Symmentec
- 3) Mimecast
- 4) Zscalar

Self-Assessment

- There were so many new things to learn from this Internship working as cloud engineer.
- After joining this internship found that each day get new thing to learn and explore more knowledge.
- Seniors and other team members help to learn it better and to solve problems.
- Learned various monitoring technologies working with site 24x7, Azure and how to process entire situation if get fail or stop working.
- Deploying new components in azure such as virtual machine, disk, Virtual network, creating storage account working with windows PowerShell script and automating the task
- learn how to get direct access to client side then problem solving it also help to more self improvement.
- 7)Learnt more new technologies like Cloudally,Zscaler,Mimecast
- working of Azure from starting from how to use to how to deploy new Vm ,
- Working with team of senior person was adding a great value in learning experience, working in an IT Organization with team was good learning curve.
- Organizing your work and put a timeline make your work easier.
- Report making, documentation & presentation are the additional skills which has been improved



S&L India IT Consulting Services Private Limited

Date: 20th Jan, 2020

To,
Sharad Gore,
Mumbai.

Subject: S&L India IT Consulting Services Private Limited - Letter of Offer

Dear Sharad,

Further to our discussion regarding your prospective employment with **S&L India IT Consulting Services Private Limited** we are pleased to extend a formal letter of offer to you. You would join the organization in the capacity of "**Infrastructure & Cloud Engineer**" and will be reporting to the "**Managing Consultant**" at this time. The scope of your job would primarily relate to but not limited to;

- Project consulting, design & implementation for S&L India's clients
- Maintain and manage customer environments for solutions deployed by S&L India and/or its related entities or otherwise depending on the clients' requirements and contracts
- You may be further required to undertake projects or activities undertaken by S&L India & S&L International and/or its related entities, and this shall be communicated to you from time to time.



S&L India IT Consulting Services Private Limited

- Income tax deduction will vary from time to time based on your investment declaration.
- LTA is not paid monthly but accrued until claimed (as per IT rules). If unclaimed, the accrued amount is paid as taxable in financial years closing month (March). It provides tax benefit and hence included as default in the structure. However, one has an option to opt out and include the same amount in Special Allowance based on preference
- The bonus, if applicable, is fixed and paid annually in the month of Diwali (Pro- Rata)

Your probation period will be six months starting from 20th Jan, 2020, or earlier depending on your actual joining date, subject to review at the end of those six months during which salary revisions would take place based on your performance.

A formal letter of appointment into the company would be presented to you upon joining.

Please confirm your acceptance by responding to this email and/or signing the letter below.

Welcome to the S&L Team!

Yours truly,
For S&L India IT Consulting Services Private Limited

Avinash Valiramani
Managing Consultant

Signed for Acceptance
Mr.Sharad Gore